Task 1: Understand LAN & WAN Fundamentals

🎯 Goal

Understand the difference between Local Area Networks (LAN) and Wide Area Networks (WAN) and how enterprises connect systems.

🔍 Explanation

Networks are broadly categorized based on **geographic scope**.

- **LAN (Local Area Network)**
  - Covers a small area (office, home, campus)
  - High speed, low latency
  - Uses switches and internal IP ranges
- **WAN (Wide Area Network)**
  - Connects multiple LANs over long distances
  - Uses routers, ISPs, MPLS, VPNs, or the internet
  - Higher latency compared to LAN

Enterprises typically operate **multiple LANs connected via WAN links**.

🛠️ Hands-On Exercise

- ➢ Identify all devices connected to your local network.
- ➢ Find your **default gateway** and determine whether it represents a LAN or WAN edge.
- ➢ Identify whether your internet connection represents a WAN link.

Task 2: Understand Networking Models (OSI & TCP/IP)

🎯 Goal

Understand how network communication is structured using layered models.

🔍 Explanation

Networking models simplify communication and troubleshooting.

- **OSI Model (7 Layers):**

Physical → Data Link → Network → Transport → Session → Presentation → Application

- **TCP/IP Model (4 Layers):**

Network Interface → Internet → Transport → Application

Each layer has a defined role, allowing engineers to isolate network issues efficiently.

🛠️ Hands-On Exercise

- Write down the 7 OSI layers in order.
- Map protocols to layers:
    - HTTP / HTTPS
    - TCP / UDP
    - IP / ICMP
- Identify which OSI layer a router and firewall primarily operate on.

Task 3: IP Addressing & Subnet Basics

🎯 Goal

Understand how devices are uniquely identified and grouped on networks.

🔍 Explanation

An **IP address** identifies a device on a network.

- IPv4 example: **192.168.1.10**
- Subnet mask defines the network boundary
- Private IPs are used inside LANs
- Public IPs are used on WAN / internet

Subnetting allows efficient IP allocation and network segmentation.

🛠️ Hands-On Exercise

❖ View IP configuration:

```
ipconfig         (Windows)
ipa / ifconfig (Linux)
```

❖ Identify:

- IP address
- Subnet mask
- Default gateway

❖ Determine whether your IP is private or public.

Task 4: Network Devices & Their Roles

🎯 Goal

Understand how different network devices control traffic flow.

🔍 Explanation

Key networking devices include:

- **Switch** → Connects devices within a LAN
- **Router** → Connects LANs and WANs
- **Firewall** → Filters traffic based on rules
- **Load Balancer** → Distributes traffic across servers

Each device plays a role in performance and security.

🛠️ Hands-On Exercise

- Identify which device performs:
    - NAT
    - Packet filtering
    - MAC address learning

- Identify your default gateway device.

Task 5: Common Network Protocols & Ports

🎯 Goal

Understand which protocols enable common network services.

🔍 Explanation

Protocols define communication rules.

Common examples:

- HTTP → 80
- HTTPS → 443
- DNS → 53
- SSH → 22
- RDP → 3389

SOC teams monitor protocols and ports to detect abnormal traffic.

🛠️ Hands-On Exercise

➔ List 5 common ports and their services.

➔ Identify which protocols are encrypted and which are not.

Task 6: DNS Fundamentals

🎯 Goal

Understand how domain names are resolved to IP addresses.

🔍 Explanation

DNS (Domain Name System) translates domain names into IP addresses.

Resolution flow:

Client → Resolver → Root → TLD → Authoritative server

DNS is a common attack vector and a critical SOC data source.

🛠️ Hands-On Exercise

➔ Run:

```
nslookup google.com
```

➔ Identify:

- DNS server
- Resolved IP address

➔ Flush DNS cache:

```
ipconfig /flushdns
```

Task 7: DHCP Fundamentals

🎯 Goal

Understand how devices automatically receive network configuration.

🔍 Explanation

DHCP assigns:

- IP address
- Subnet mask
- Gateway
- DNS servers

DHCP process: **DORA** (Discover, Offer, Request, Acknowledge)

🛠️ Hands-On Exercise

- ❖ Release and renew IP:

```
ipconfig /release
ipconfig /renew
```

- ❖ Observe lease details.

Task 8: Routing & Traffic Flow

🎯 Goal

Understand how traffic moves between LANs and WANs.

🔍 Explanation

Routing determines how packets move across networks.

- Routing tables define paths
- Default route sends traffic outside the LAN

🛠️ Hands-On Exercise

1️⃣ View routing table:

```
routeprint(Windows)
iproute(Linux)
```

2️⃣ Identify:

- Default route
- Gateway IP

Task 9: Network Troubleshooting Tools

🎯 Goal

Learn tools used to diagnose network issues.

🔍 Explanation

Common tools include:

- **ping** → Connectivity
- **tracert / traceroute** → Path analysis
- **netstat** → Active connections
- **arp** → IP–MAC mapping

🛠️ Hands-On Exercise

➔ Test connectivity:

```
ping google.com
```

➔ Trace network path:

```
tracert google.com
```

➔ View active connections:

```
netstat -ano
```